INTENTIONALITY, TEMPORALITY, CORPOREALITY AND ORGANICISM IN LIVE ELECTRONICS

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ABSTRACT

This paper responds to John Croft's critique of compositional approaches to works for instrument(s) and live electronics, as presented in his article "Theses on Liveness"[1]. Backgrounded by a discussion of the metaphor of "organicism" in music, and its extension from purely formal considerations to include the nature of the relationship between performer and computer, this paper teases out further limitations of the medium: namely, an inherent *intentionality imbalance*, an *arrow of causality* and an innate *temporal lag* between performer and computer. Furthermore, if we agree with Croft's diagnosis of much of live electronics' lack of intrinsic "difficulty"—in the sense of the kind of physical labour seen as integral to the experience of live music—then the only route left open is to accept the role of the computer as passive and insentient. Drawing the analogy to both models of passive instrumental design (e.g. the Aeolian harp) and spatial forms (e.g. the reverberant cathedral), this paper proposes a number of possible models that extend beyond Croft's preferred 'instrumental' paradigm, into areas that touch more upon sound installation praxis, to provide 'future work' for composers and sound artists who are concerned to uphold the ontological necessity for retaining the corporeality of the live experience, whilst avoiding the pitfalls of triviality.

1. INTRODUCTION

This paper concerns aesthetic issues that arise in the composition of music for solo instrument (or instrumental ensemble) and live electronics. For the purposes of this paper, I will limit myself purely to discussion of the context in which the live electronics are *not* being controlled by an on-stage musician with a physical, performative presence—which may, *potentially*, mitigate some of the objections rehearsed here—but is instead processed by a device (which I will call, for ease of reference, a "computer") with minimal, or at least unobtrusive and non-performative intervention from either a technician or the performers themselves.

Within this scope, I argue that the dominant paradigm of composition is, in essence, an extension of the late eighteenth-century musical metaphor of "organicism". In live electronics, this approach incorporates not just the formal processes of material proliferation and part/whole relationships, but also encompasses the sonic and interactive relationship between live sound and electronics. The electronics carry much of the work of proliferation and variation, and maintains close sonic ties to the performer's sound material, with allowances for sonic departures as a developmental strategy. It has become the dominant paradigm not just because of historical and cultural exigencies, but also, I argue, because of two inherent problems with the very concept of live electronics: firstly, there is an intrinsic *arrow of causality* from musician to computer; secondly, the two parts of the live electronics equation—human and computer—possess, and clearly display, an inherent *imbalance of intentionality*.

Live electronics systems have an inherently fixed linear temporality from human to computer precisely because *both* the computer is non-sentient *and* the performer is sentient. We accept, even before a single sound of the piece has been heard, that the computer is not "truly" responding, in any sense that might be called intelligence. But any response is intrinsically infallible and high fidelity. On the other hand, we accept immediately that the human is intelligent, whilst being highly fallible.

This imbalance of sentience and fallibility is perceived by the listener as an imbalance of *intentionality*. The performer is always perceived as the *initiator* of any discursive act, for an *intention* must be generated by them for any musical act to take place. Furthermore, there is always the chance that the performer's fallibility may render them mute (thus every musician's nightmare of picking up their instrument for a solo, and being unable to make a note). This fallibility further heightens the tensional aspect of the live experience.

Meanwhile, the computer is perceived as an insentient, passive *responder*. It is worth comparing this situation to the experience of listening to purely acousmatic music, in which we do not try to

[1] Croft (2007)

anthropomorphize the technological agency: we hear the music as a form of 'sonic document'. Without a corporeal presence, there is no imbalance to be resolved.

A good analogy for live electronics is that of the Aeolian harp, an instrument that responds unthinkingly, but not unbeautifully, to the wind passing through it. But a performer cannot engage in a meaningful and truly contrapuntal musical discourse with an Aeolian harp, not even by blowing *really* hard on it: its passivity and lack of intentionality prohibits this[1]. Its passivity, however, is also its beauty, and we can take some measure of aesthetic pleasure from the way it sonifies changing wind currents. I will return to this point later.

The imbalance of intentionality causes a *fixed linear temporality* and *arrow of causality*: the computer must always temporally *lag* the performer, or, at best, shadow the performer. It seems difficult to conceive of a situation in which this is *not* the case, in which we can say that the computer is truly *preceding* the performer. I will also return to a finer-grained discussion of this point later.

Organicism in live electronics, then, accepts as an *a priori* fact this fixed temporal disposition of the live electronics. In fact, it not only accepts it, it exploits it. In organic works, both performer and computer have well-defined roles: the performer acts as the *germinator* of material, which the computer *proliferates*, typically using processes involving variable temporal delays and variation techniques to create a web of material permutations, in the spirit of conventional developmental strategies.

In the article "Theses in Liveness", composer John Croft provides a substantive aesthetic critique of the field. He questions whether most of the dominant paradigms in live electronics are a symptom of "technology fetishism" rather than poetic considerations. I contend that while, in many ways, I believe Croft's objections to have a *prima facie* validity, I do not believe that organicism in live electronics is therefore dead. But it asks of us to maintain a critical approach to the relationship between performer and electronics, particularly in the conception of a live electronics practice that does not fall foul of "shifting the procedural into the foreground", thus becoming "a lamentable 'showcase' of the technology".[2]

2. THE METAPHORICAL ROOTS OF ORGANICISM

The metaphor of organicism arose in the late eighteenth century as concepts of growth and biological and physiological development in the physical sciences were adopted by music critics to replace the earlier rhetorical models of musical discourse. That is not to say, however, that a single, consistent metaphor was arrived at: indeed, there are competing metaphors of organicism that focus on different aspects of the musical work as simulacra of a biological organism. The most prominent and persistent metaphor, however, holds the image of the temporal unfolding of material as a *proliferation* and *variation* of autonomous musical units following accretive and degenerative processes. It could be seen to draw analogies with the processes of cell division, mutation, DNA encoding, natural replication errors and cell death that provide the underlying processes by which biological organisms grow, vary and decay[3]. The central image, however, is that of the concept of a germinating 'cell', which contains within it the recipe for its forthcoming formal organisation, as it sends out and accretes branches, rhizomes and inflorescences to the point where a singular, coherent whole is perceived. To quote nineteenth-century music critic Adolph Bernhard Marx, the motif is "the germinal vesicle, that membranous sac filled with some fluid element…the *Urgestalt* of everything organic."[4]

This metaphor owes its aptness to the cognitive faculties of the brain, particularly its ability to form, and maintain within memory stores, perceptual Gestalts. It can recognise cognitively pertinent feature-types, such as contour and interval of prominent motivic units, and track changes and accretions to these features over time. The metaphor of organicism, through the innate drive towards proliferation and variation extraneous of any sacred or secular function, arises almost contemporaneously with the notion of autonomy in art, which enters the music literature in about 1790, only 20 years after the concept of organicism first appears.

3. THE HISTORICAL AVANT-GARDE AND ITS DECONSTRUCTION OF ORGANICISM

The dismantling of the privileged position of musical organicism came most concertedly with the radical overhaul of materials, techniques and processes that accompanied high modernism. Although serialism was conceived as a fundamentally organic strategy, the limits of human Gestalt-forming abilities were, arguably, breached. It is ironic that Schoenberg ended up having to rely heavily on conventional non-pitch features, such as rhythmic character, texture, register and approximate contour, to convey a sense of familiarity, which the row itself—which was supposed to have been the 'germ'—was unable to provide. With the advent of total serialism in the early 1950s, music achieved the most unusual of situations: a piece in which virtually every notated parameter derives from a generative rule-set, not unlike the construction of a complex organism, but strangely ending up with what many commentators have dubbed as "inorganic".

[1] This partly explains why there are not many Aeolian harp concertos in existence.
[2] Croft (2007), 61

[3] Even though many of these processes were unknown to science at the time of the metaphor's genesis.
[4] Marx (1856), quoted in Spitzer (2004), 291.

Stockhausen's famous description of Messiaen's *Mode de valeurs et d'intensités* as "star music", is one example, in which the inherent stasis and lack of conventional gesturality requires reference to something extraterrestrial.

For a genre as technologically progressive as live electronics, it seems surprising that most works do not engage more wholeheartedly with inorganic formal constructions. But in the light of the inherent temporal and causal limitations of live electronics, not to mention the technological ease of certain modes of audio processing, perhaps it is not surprising. After all, the construction of a situation in which inorganicism dominates proceedings does, in fact, take on not just a conceptual disinclination but also a technological obstacle. The works of Karlheinz Stockhausen perhaps stand out as exemplars of an "inorganic" live electronics practice, though with the qualifier that his works predominantly rely on synthesis, which has the advantage of not being implicated so clearly in the one-way causality I mentioned earlier. I will return to questions of inorganicism later; for now I want to turn to a recent critique of the current dominant practice within live electronics.

4. CROFT'S THESES ON LIVENESS

John Croft's article "Theses on Liveness" is a provocative assessment of the current status of music for instrument and live electronics. It provides an intellectually robust critique of the common practice in live electronics, arguing for the necessity of maintaining a musical ontology that values corporeality as an essential part of the aesthetic experience.

If one agrees with the premises of the argument, then the conclusions are profound for a live electronics practice. If organicism in live electronics extends to the *relationship* between the acoustic and electroacoustic components, then Croft's taxonomy of relationships attempts to reduce these relationships to five basic types, of which at least three would be necessarily organic:

- *Backdrop*: the electronics serve primarily as a ground on which the instrumentalist provides a figure. There are no perceptually causal relationships between the background sound and the live sound.
- *Accompanimental:* the electronics server primarily to accompany the instrumentalist in a conventional sense, perhaps with the aid of triggering or score-following.
- *Responsorial/proliferating:* the electronics have an antiphonal relationship to the live sound.
- *Environmental:* the electronics simulate aspects of a virtual environment.
- *Instrumental:* the electronics combine with the live instrument sounds to create a "composite instrument".

Croft argues that not only is there an imbalance in intentionality between the two components, but also that we respond to a physical system differently through our empathising with physical labour—the ease or difficulty with which any given sound might be produced—and with the psychological tension of fallibility. Croft spells out this position clearly: "A loudspeaker can, in principle, produce any sound; on an instrument, almost all sounds are impossible, and of those that are possible, some are more difficult to produce than others, and this difficulty is patent in the act of performance". He continues: "This is surely why performance engages us in a way that cannot be accounted for in terms of the sound alone: the difficulty, the impossibilities, the encounter with limits, the *finitude* of the instrumental performance resonates with wider human experience."[1]

For Croft, the labour and precariousness of our corporeal reality is *stylised* through live instrumental performance. A listener's innate tendency to empathise with struggle, the overcoming of the odds, and to marvel at the mastery of such fallible systems is crucial to an aesthetically valorised experience.

The very real possibility of electroacoustic sound being just *too easy* and too divorced from physical labour strikes at the very understanding of music. As Croft points out, it suggests that there are two competing ontologies of music: that of music as irreducible from its sound, and that of music as inseparable from its physical conditions; as he puts it: "from the intersubjective and social act of performance, with the significance of effort, labour and expenditure, and indeed with the erotics of physical engagement."[2]

5. WHERE TO FROM HERE?

Croft's taxonomy seems, on the face of it, to be highly proscriptive to the composer. It might seem to shut the door on the majority of approaches to the genre, a position that might be construed as an *a priori* critique of the entire field.[3] But it is my contention that there are sufficient byways, tributaries, and fertile backwaters within live electronics that still adhere to Croft's preferred ontology for composers to continue to develop compelling and meaningful. What I propose below is therefore a sketch for further work: an outline of potential fields that are ripe for exploration and refinement, that hold true to the essential corporeality of the live experience, but in which the infallibility and fidelity of the electronics may be exploited in a non-trivial way.

[1] Croft (2007), 62
[2] loc. cit.
[3] Which, given he is an accomplished composer of music for instrument and live electronics, would be largely unfair.

5.1. Memory nets

I would first like to suggest that the ability of the computer to accurately record and reproduce live audio with great fidelity is the primary strength of live electronics, and any system that uses this to its strategic advantage in a thoughtful and non-trivial fashion may well go some way to overcoming Croft's objections. I suggest a new paradigm here, that of a "memory net". In a common live-electronics setup, sound is recorded and immediately *recycled* typically through some sort of time-domain brassage or delay process, which leads to a *texturisation* or *fragmentation* of the audio stream. In Croft's taxonomy, this is the *responsorial/proliferation* paradigm. As mentioned, this is the most commonplace approach to live electronics: it is relatively simple to implement such a system without much work. A more artistically potentiated approach, however, is to store incoming audio in a buffer and recall it at a much later point in the piece in such a way that it will fulfil a considerably different function than mere texturisation. Digital audio can compellingly, even shockingly, recall exact moments in time, and its recollection might constitute either a *summation* of past events or even the *disruption* of current events with a dispute from the past. A poetic example of this is provided by Karlheinz Essl's *Entsagung* for chamber ensemble and live electronics, in which live instrumental gestures from the opening of the piece are stored away and return at the end in an increasingly dense texture, whilst the live players drop out of the score completely. The psychological impact is compelling: we perceive the electronics as a kind of distorting mirror of the past, re-folding prior events into the ever-present now. The re-presentation of the past presents the listener with a situation we are rarely presented with in everyday life, for it has neither the role of an echo—which we take to be causally linked to a sound event that occurred in the very recent past, typically no more than 10 seconds ago—nor as archival footage, which we take to have occurred in the more distant past, anywhere from a few days ago to centuries ago. Instead, the replaying of audio from 5 or 10 minutes ago has the effect of *déjà vu*, a disorientation of the perception of now.

An objection is that this form of aural recollection has been around for centuries, in the form of recapitulatory structural processes in instrumental music. In this case, however, the re-appearance of past events is *allusive*, in the sense that two people talking about a past event *report* what happened, rather than literally replaying the occurrence verbatim.

With live electronics, however, the past incurs on the present with shocking fidelity, in a way that *cannot* be merely allusive, for its very literality acts as a sonic document.

5.2. Dominance/subordination

The formal process at the end of Essl's *Entsagung* suggests another paradigm in live electronics, that of the persuasive manipulation of the *hierarchy* between performer and computer, between that of dominance and subordination. In conventional works, the live performer is active for most of the piece, due to the composer's necessity to maintain the physical presence throughout. As such, the instrumental part often could be played without the electronics without significant aesthetic loss.

But we might conceive of an instrumental part less as a potentially standalone piece, and give space for the computer to emerge as a dominant discursive voice at times throughout the piece. It may very likely require the performer to drop out altogether: the difficulty here is the composer's aesthetic prerogative to convince us that this is a meaningful and musically enriching relationship: the computer must significantly vary and direct the discourse, without the crutch of the live sound.

5.3. Non-linear temporality

Aside from the *recollection* of events past, live electronics also has the potential to develop non-linear temporal structures beyond the conventional teleological "unfolding". Recent (and even not-so-recent) literary and filmic developments have questioned the necessity for linear narrative; but in conventional instrumental works, with the abstractness of its musical material, it is very difficult to achieve the illusion of non-linearity, as cultural conditioning and, potentially, cognitive constraints lead the listener to infer a linear organic unfolding even where one was not actually present in its construction. In other words, we can only rarely stop ourselves from hearing a beginning, middle and end, and strive valiantly to relate the sound material to each other.

In live electronics, however, the role of literal sonic recollection has the possibility to produce a much more non-linear presentation of material. Material that, in a linear presentation, would *precede* other material can be presented later in the work and contextualised with a literal representation of the later material. Again, the composer faces the task of convincing a listener of this non-temporality, and it remains an open project as to how true non-linearity can be achieved.[1]

5.4. True contrapuntality

If the implicit linear temporality of the relationship between performer and electronics—the "fixed imitative canon" I alluded to above—provides us with conceptual pause for thought, then a potential development is to give considerable weight to finding a technological and conceptual solution to the diversification of contrapuntal interrelationships. Implicated in this is the intractable notion that the composer must posit a situation in which the performer may truly be able to respond to or delay the electronic

[1] Significant work by composers such as Lindsay Vickery is progressing the possibility of non-linearity in these domains.

part. For this to be perceived as a true *response*, however, the audience must be able to perceive the *intentionality of response* on the performer's behalf. A conventionally notated score would seem to proscribe this possibility: even if the response and delay is written into the score, the perception is still that the performer is following the score, not the electronics. An objection might be that in a conventionally notated instrumental contrapuntality, performers do not *really* respond, at least in any improvisational way, but we must not forget that in this case there is no imbalance of sentience between members of, say, a string quartet, viola jokes notwithstanding. It might also be objected that it is easily possible for live electronics to generate material that precedes a response from the performer; this returns us to the *accompanimental* category of Croft's taxonomy, and we perhaps return to asking whether this would be better served as an instrument-and-tape paradigm. With an open score, or improvisational practices, it may be more conceivable to posit a situation of true interrelationship, but, even so, it may yet prove elusive, and it remains an open question, as to whether we can ever develop a *truly* two-way, interactive relationship between human and computer.

5.5. Microsound

Croft rightly cites Nono's pioneering works in live electronics, *Das atmende Klarsein* and *A Pierre* as exemplars of music that is predicated on the poetic implications of accentuating the transitory, fragile and vestigial in sound. It is, as Croft put its, "the grain, the 'imperfections', the unrepeatable" which constitute "the reason for the continued importance of performance". Live electronics has an almost unprecedented possibility of drawing to our perceptual attention those elements of sound that are typically of ancillary or even non-existent cognitive focus. These elements may be highlighted not only purely through the role of electronic amplification, but also through processual algorithms aimed at the analysis, decomposition and resynthesis of sound.

5.6. Sonification and space

As I mentioned at the outset of this paper, an Aeolian harp responds unthinkingly, but beautifully, to varying wind currents. It reifies the ephemeral fluctuations of air pressure, through a modal transmutation: it *sonifies* our environment, it gives voice to the ineffable. We do not, however, criticize an Aeolian harp for its inhumanity and infallibility. Rather, we might draw aesthetic pleasure from the way that different meteorological conditions give rise to very different 'voices' within the one physical system.

With live electronics, we have the potential to exploit the *sonifying* quality of a fixed system that the computer might provide. Already Croft hints at this, with regards to Nono's reinforcement of transitory aural phenomena, but I think we can take this further. We might think of other ways in which the performer might contribute 'input', other than purely sonic means. In Ruth Anderson's work *Centring* (1979), audience members are measured for their galvanic skin response which feeds the frequency of a bank of sine-wave oscillators. An on-stage dancer moves to the resulting audio, and, in theory, the audience members further respond to the dancer, creating a feedback loop between sound and stimulus. As a work of purely sonic art, it perhaps lacks the compelling sonic surface that would further transform it into a more sophisticated formal disposition—but it does provide a suggestion for further work that certainly asks us to interrogate the blurred boundary between mixed live electronics works and sound installations.

By expanding the concept of sonification beyond the dimensions of the object into the territory of *spaces*, however, we can utilise the insentience of the computer in a manner analogous to the acoustic response of a spatial environment. Not only can we simulate a virtual spatial environment, which is, perhaps, the least interesting version of this concept, but we can also interact with an existing physical space in ways that would be impossible with live performers. As a thought experiment, we might conceive of a work housed in a large resonant acoustic, such as a cathedral, in which the single performer generates subtle, almost inaudible sound which is, however, close-miked and proliferated out to multiple loudspeakers distributed as widely as possible throughout the space. As audience members pass through different parts of the cathedral, the changing resonances, changing processing and changing sonic impetus from the performer would provide a much more complex and engaging spatial encounter — most importantly to this concept would be the sense that it is not possible to hear all parts of the "piece". Like the stone garden at Ryoanji in Kyoto, in which you can never see all of the "rock islands" at once, in this piece there would always be sound occurring that would be unable to be heard by any one person.

5.7. Orchestral

Taking this idea of a spatially-oriented electronics further, we might conceive of a situation in which a massively multi-loudspeaker setup exists within a large acoustic, in which the organicist conception is taken to an *orchestral* extent. Through the development of sophisticated microvariation techniques, multiple microscopic delay offsets, and the storage of an intelligent recall of sound input, the composer could feasibly generate the kind of fine-grainedness that we appreciate in a live orchestral performance. Part of the joy of attending an orchestral concert is the possibility of attending to individual voices; in this case, massively multiple outputs would be required, each routed through a separate loudspeaker, to "contribute" in a heterophonic manner to an overall impression of global form. With sufficient computing and programming sophistication, the "orchestra" could

even be "sectionalised", with different loudspeaker subsets — "choirs" in orchestral parlance — taking on different material in a stratified disposition.

The potential within this idea for macrosructural formal processes to be articulated through spatial disposition is hinted at by Janet Cardiff's *Forty-part Motet*, in which Tallis's *Spem in Alium* is spatialised through 40 separated loudspeaker channels. In this work, the *tutti* sections take on a heightened perceptual significance, to the point where the sudden entry of all 40 voices surrounding the listener contributes to a heightened affect.

5.8. Inorganicism, textuality and extrinsic referentiality

Finally, we might conceive of a musical work that features inorganicism in its formal construction, emphasising disjuncture, fragmentation, chance operations, and the deliberate fracturing of linear growth and decay trajectories. *Or* we might focus our attention on the organic relationship between performer and computer. Yet a simplistic, deliberate mismatch between electronic sound and live performer probably only returns us to the same paradigm as instrument with tape: the *environment* category in Croft's taxonomy, with its attendant degenerate case of rainforest flute.

One solution is to conceive of a relationship in which, rather than the typically unquestioning acquiescence to and subsuming of the live sound by the computer, instead there is a *provocation* by the computer. The problem of one-way intentionality still limits the degree to which provocation can be heard as truly a performative act on behalf of the computer. But one area that might be worthwhile to explore is in the area of vocal textuality, in which the extrinsic referentiality of recorded text might fruitfully interplay with live text.

Indeed, extrinsic referentiality, especially in the form of historically-encoded sound sources or other sounds that resonate with wider sociocultural contexts, may well continue to present compelling artistic forms in which the relationships of live electronics are still

6. SUMMARY

If we accept the cautionary tone of Croft's writing, and I contend that there are indeed artistic and poetic reasons to attend to his injunctions, then I think we are faced with a choice: to either reject live electronics as, *a priori,* a dead end—which indeed many composers do—or to accept and indeed even exploit its inherent imbalances and limitations, in order to conceive of pathways for future work.

I argue that forms which accept and exploit the inherent insentience and passivity of the computer, in a manner perhaps more akin to sound installation, have a better chance of generating new musical situations than the conventional contrapuntal method.

Of course, in the end, we will just compose the music we want to compose, but attendant to the creative urge should always be the question of "why?" And if a search for answers energizes our creative urge, rather than diminishes it, then such discourse is not without its place.

7. REFERENCES

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